CARRONADE

With a sonic boom, clear the gloom, Carronade makes your harvests bloom.

Complete Description

Carronade is an **innovative, automated, and eco-friendly device** designed to protect agricultural fields from pests using **sound-based deterrence**. It operates through a controlled chemical reaction between **calcium carbide and water**, producing acetylene gas that is ignited with an electric spark to create a loud sound that scares away birds, animals, and insects without causing harm. A timed dispensing system, managed by a servo motor and water pump, ensures precise chemical mixing, while an electric ignition unit clears residual gas and ensures reliable combustion. **A motion-detecting camera with night vision** activates the system only when pests are detected, optimizing energy use and reducing noise pollution. The entire operation is managed by a microcontroller, allowing **fully automated, low-maintenance functioning**. Melem MOF sheets are used to absorb excess acetylene gas, ensuring environmental safety. Built with affordable components, Carronade is **scalable to different farm sizes and suitable for solar or battery-powered operation**, making it ideal for rural areas. Overall, Carronade offers a **sustainable, cost-effective, and non-lethal solution** for modern farm protection.

Background Of The Problem

In recent years, **farmers across India and other agricultural nations have faced increasing challenges** in protecting their crops from invasive animals and pests. Birds like crows and parrots, large animals such as monkeys and wild boars, and swarming pests like locusts often destroy fields in a matter of hours. These pest attacks directly **reduce yields, damage standing crops, and lower farmers' income.** For small and marginal farmers, even partial losses can affect their financial stability for the entire year.

The **problem intensifies during harvest seasons,** when crops like maize, wheat, paddy, and fruits become most vulnerable. Birds peck at the grains, monkeys uproot plants, and locusts devour entire fields. These **attacks are not isolated; they happen frequently and affect large areas, leading to mass-scale losses**. Often, such attacks go unchecked due to the l**ack of effective and affordable protection methods**, especially in remote or rural areas where farmers cannot access high-end technologies. The increase in human encroachment, deforestation, and destruction of natural habitats has pushed wildlife closer to human settlements, making crop fields an easy target for survival.

As a result, the f**requency of such incidents is increasing every year, creating a pressing need to address this agricultural crisis through better protection strategies.**

Farmers have been relying on various **traditional and semi-modern techniques** to safeguard their fields, but most of these are either **outdated, inefficient, or unsustainable in the long run**. The most common method—using scarecrows—has lost its effectiveness. **Birds and animals quickly adapt and learn that these figures pose no real threat**, ignoring them completely after a few days. Similarly, hanging shiny objects or loudspeakers on timers often creates constant noise pollution without offering reliable results.
Chemical pesticides and repellents are another frequently used method. While they might deter some pests, they come with **harmful side effects**. Excessive use of **chemicals affects soil fertility, contaminates groundwater, harms beneficial insects like bees, and can even leave toxic residues on food crops.**

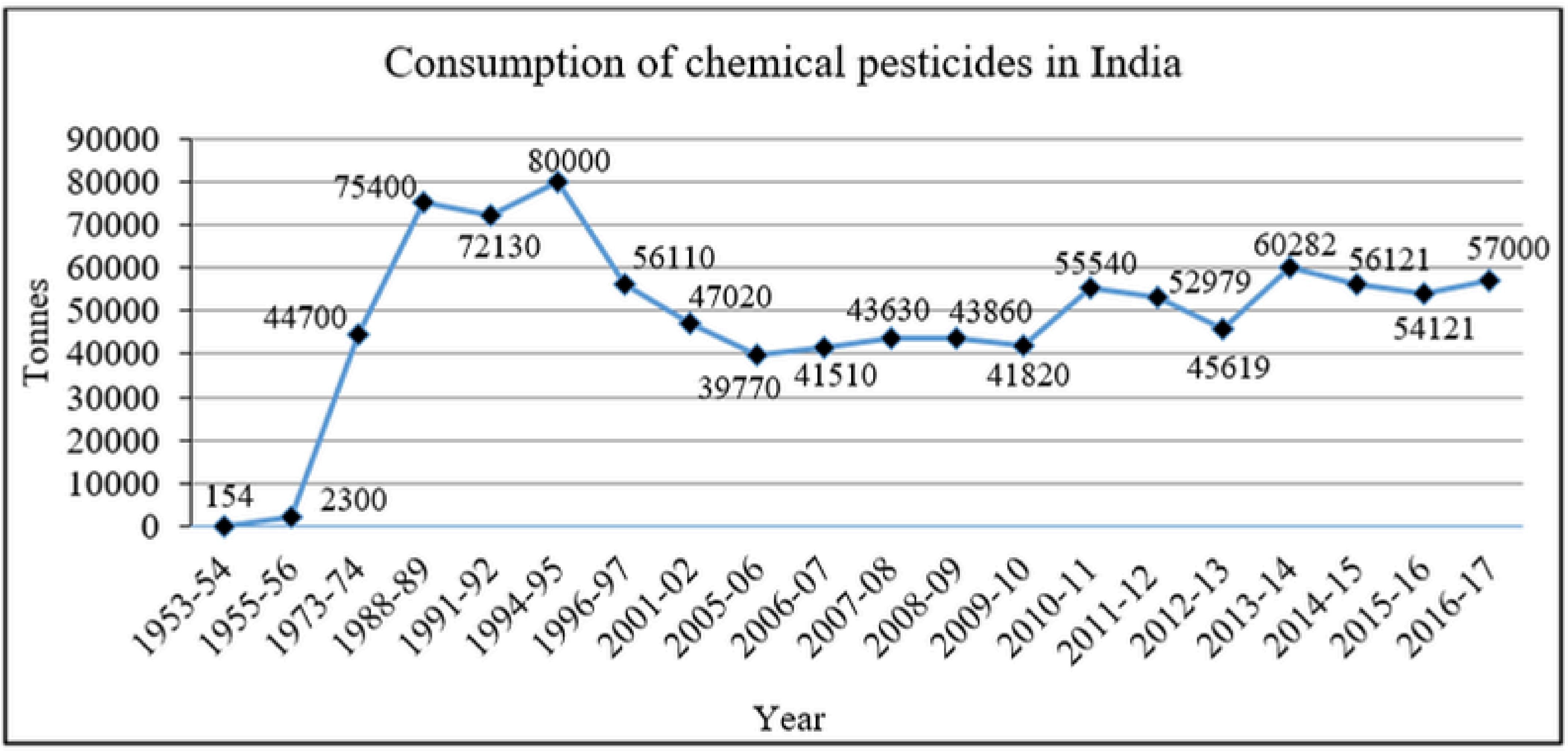
Furthermore, wild animals are not affected by such chemicals, rendering them ineffective against monkeys or boars. Electric fences and ultrasonic repellents are considered modern solutions, but they are often expensive to install and maintain. These systems also raise safety concerns, especially when they **pose a threat to animals or humans who may accidentally come into contact with them.** Moreover, such systems require a constant electricity supply, which is **unreliable or unavailable in many rural or off-grid regions.** As a result, farmers are left with few practical, safe, and affordable options.

Another major issue in current pest-control methods is the **lack of automation and smart response.** Most systems operate continuously or on fixed timers, regardless of actual pest presence. This results in **wasted energy, noise pollution, and ineffective protection,** especially in the absence of real-time detection. Farmers are often forced to **monitor their fields manually at odd hours**, losing sleep and increasing labor efforts just to keep pests away.

Night-time pest attacks are particularly damaging, as many animals, such as **wild boars or nocturnal birds, operate under the cover of darkness.** Most traditional systems fail during this period due to the absence of lighting, sensors, or visibility. Farmers often resort to burning fires or making noise themselves to guard fields at night—**efforts that are exhausting, temporary, and unproductive.**

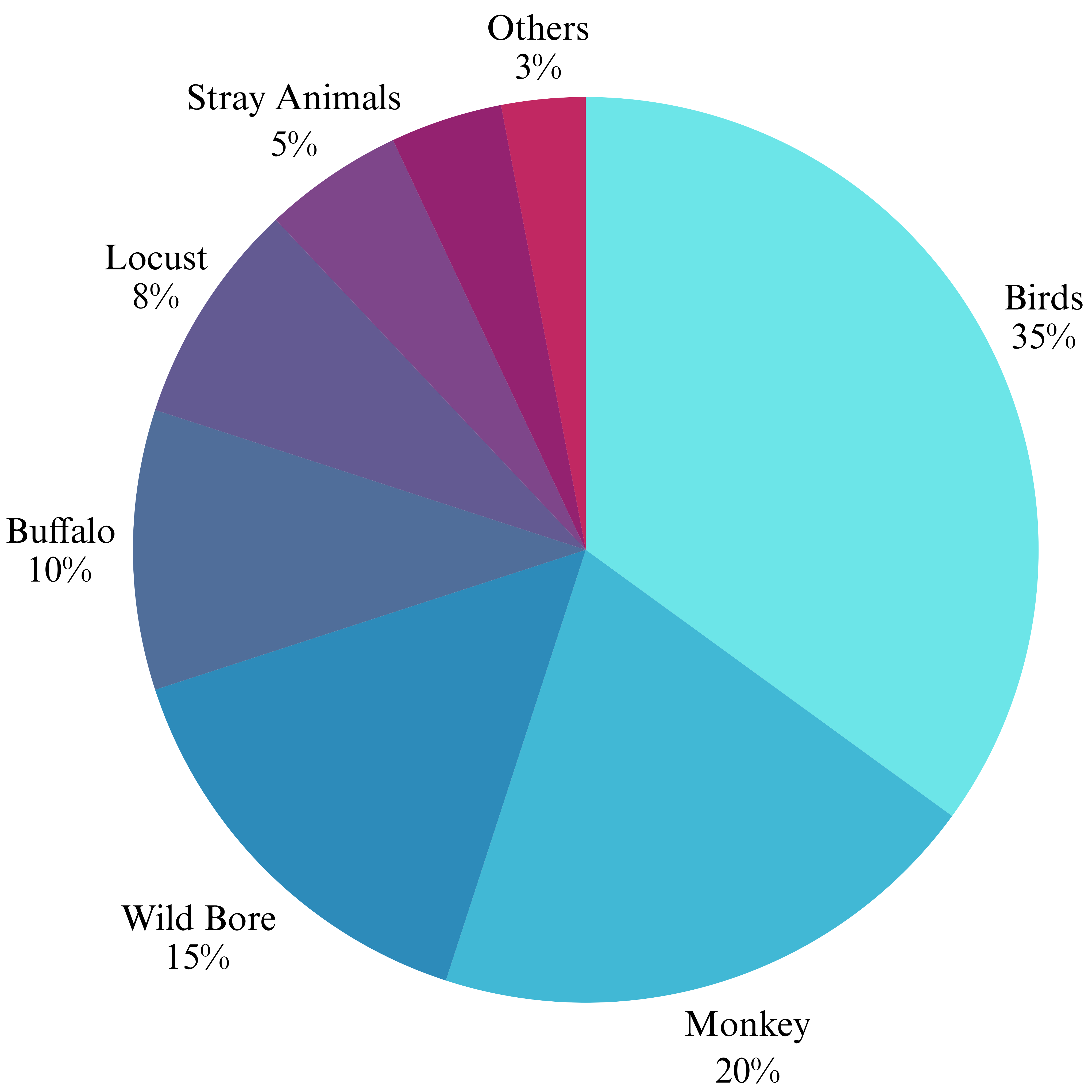
Adding to this is the financial burden. Most small and medium-scale farmers cannot afford complex pest-repelling technologies, nor do they have the technical knowledge to operate or maintain them. Language barriers, lack of training, and the high cost of maintenance make even the best technologies out of reach for those who need them most. All these factors combined contribute to a **widespread, deeply rooted problem** that continues to **affect food production,** farmer well-being, and rural economies across the country.

Uses of Harsh and Dangerous Chemicals



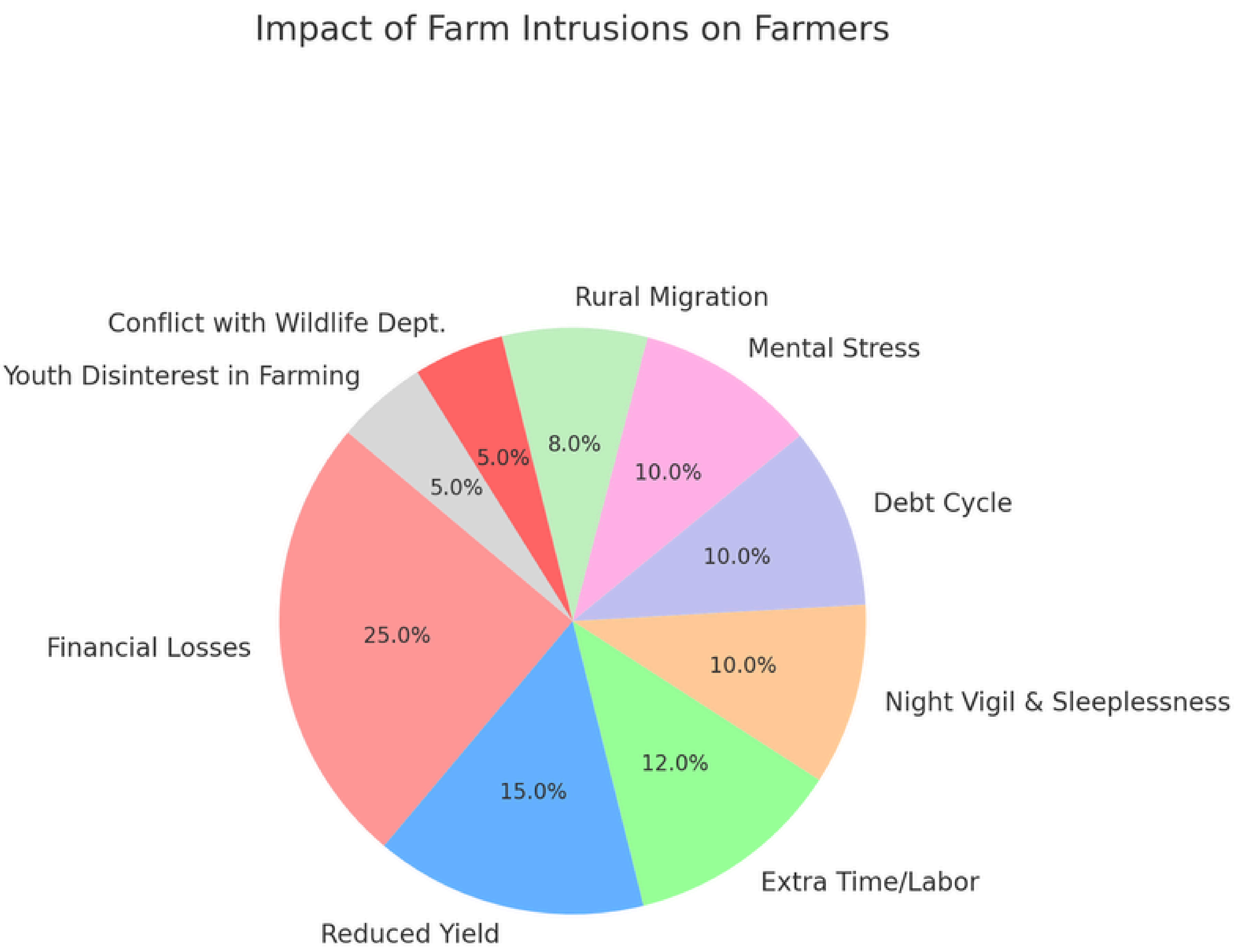
The data above results show that to remove intruders from the farm, farmers use chemicals and pesticides to protect their harvest. These methods remove the intruders but damage the crop and harsh chemicals can further harm the health of the consumer.

<u>Type Of Intrusions In The Farm</u>



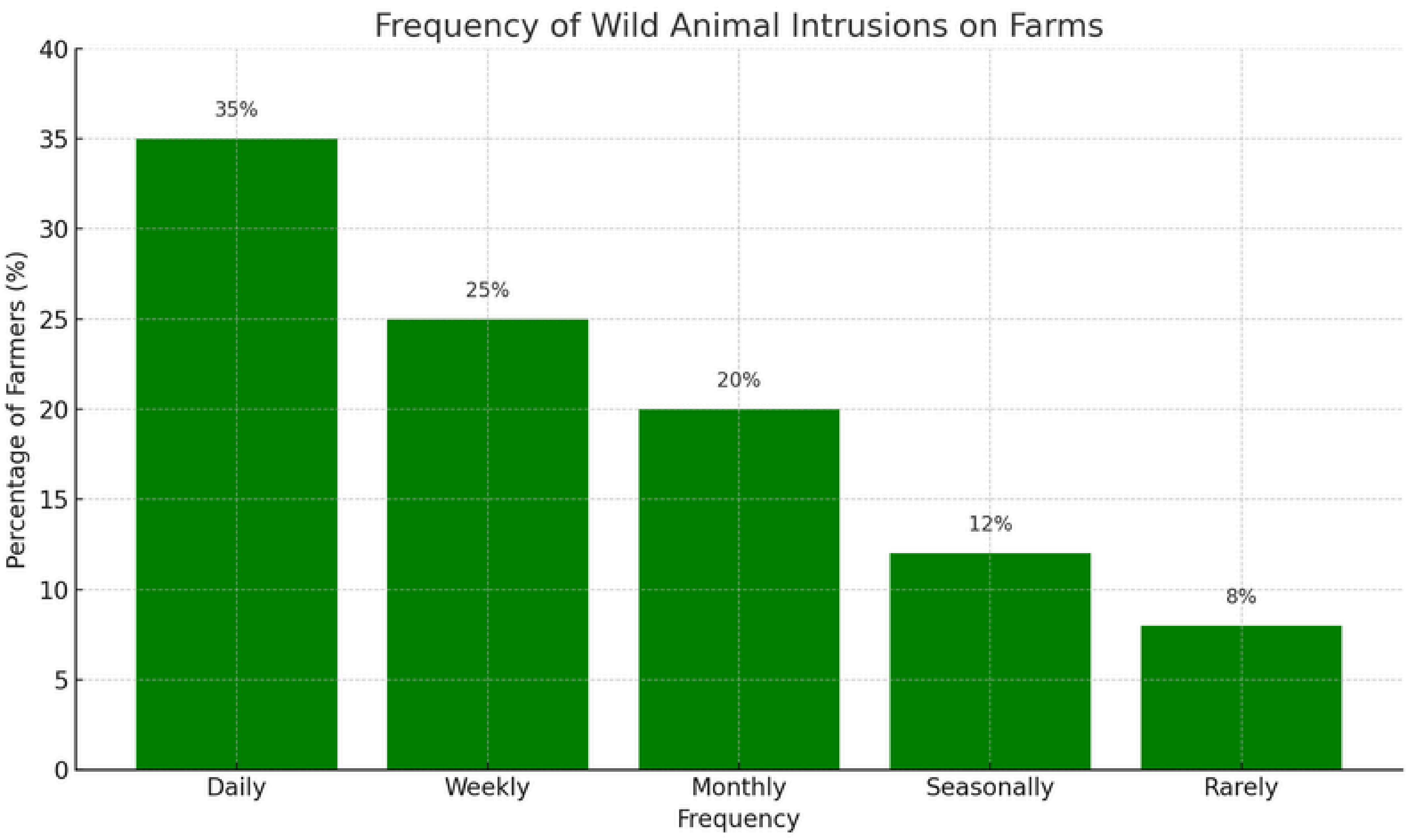
The data shows which intruders enter the farm and the percentage of intrusions

Problems Faced by Farmers During Intrusions



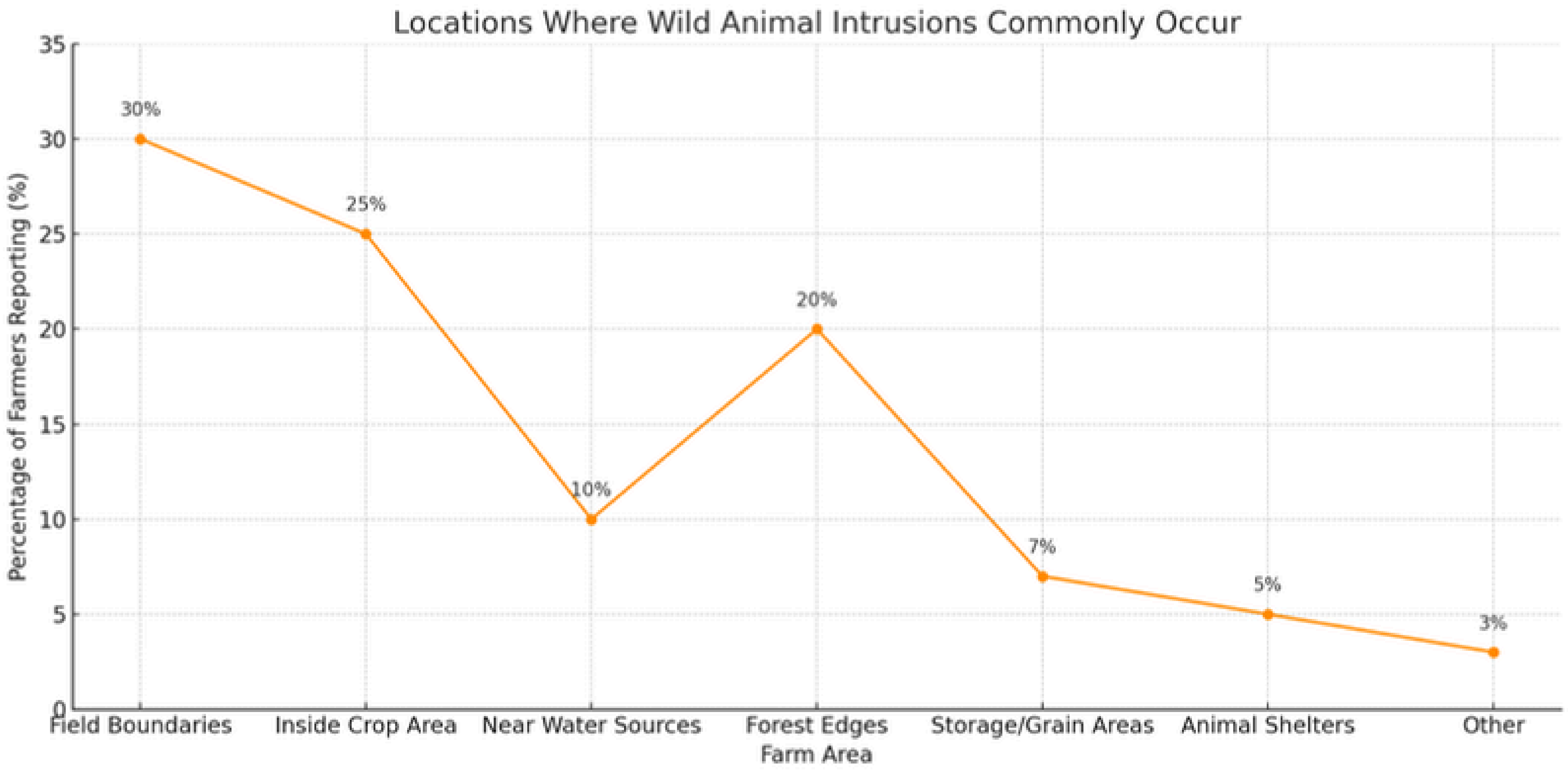
The results show that farmers face many anxiety and problems when their crops are ruined

Type Of Intrusions In The Farm



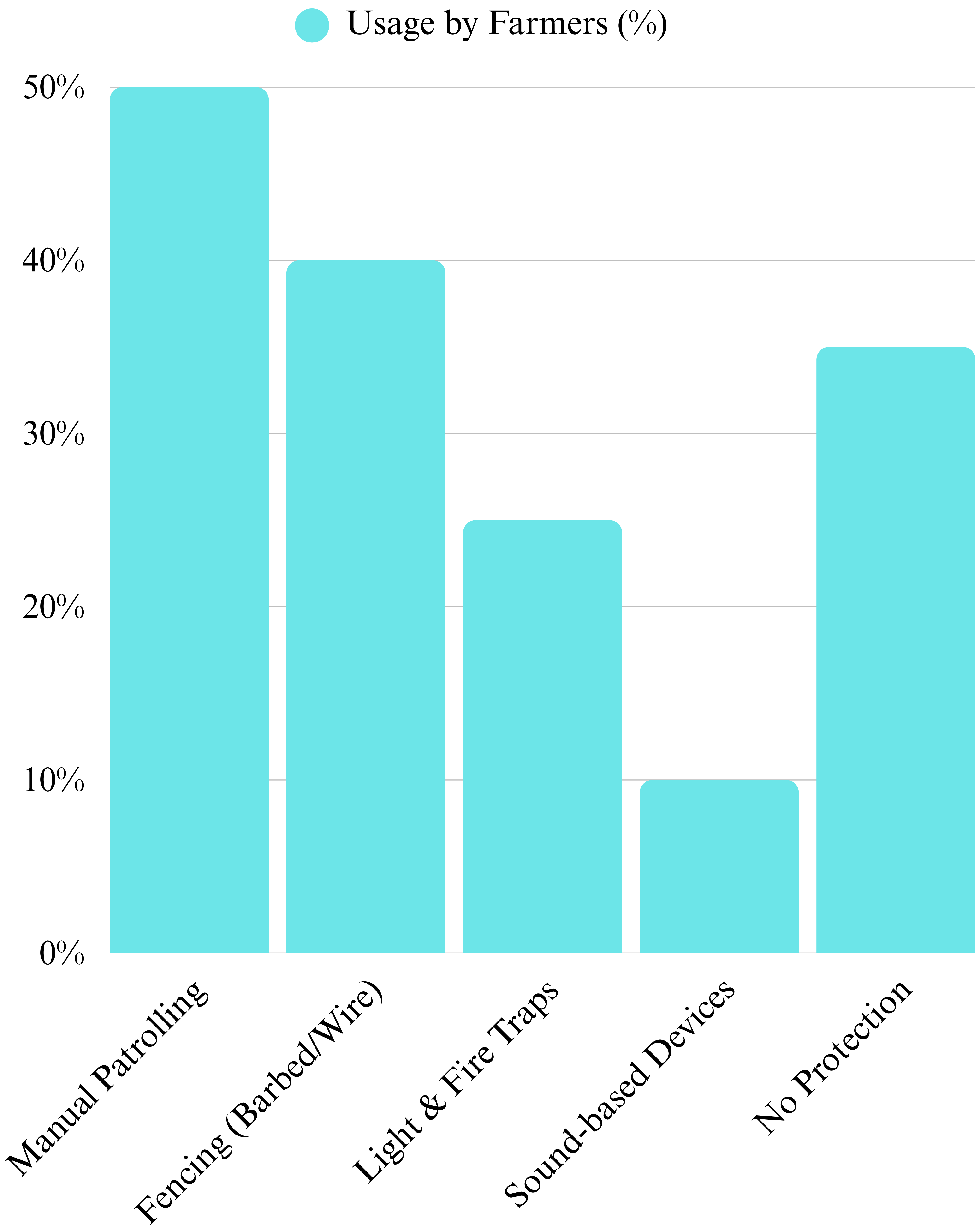
The data shows how frequent farmers experience intrusions in their farms

Location Where Intrusions Occur



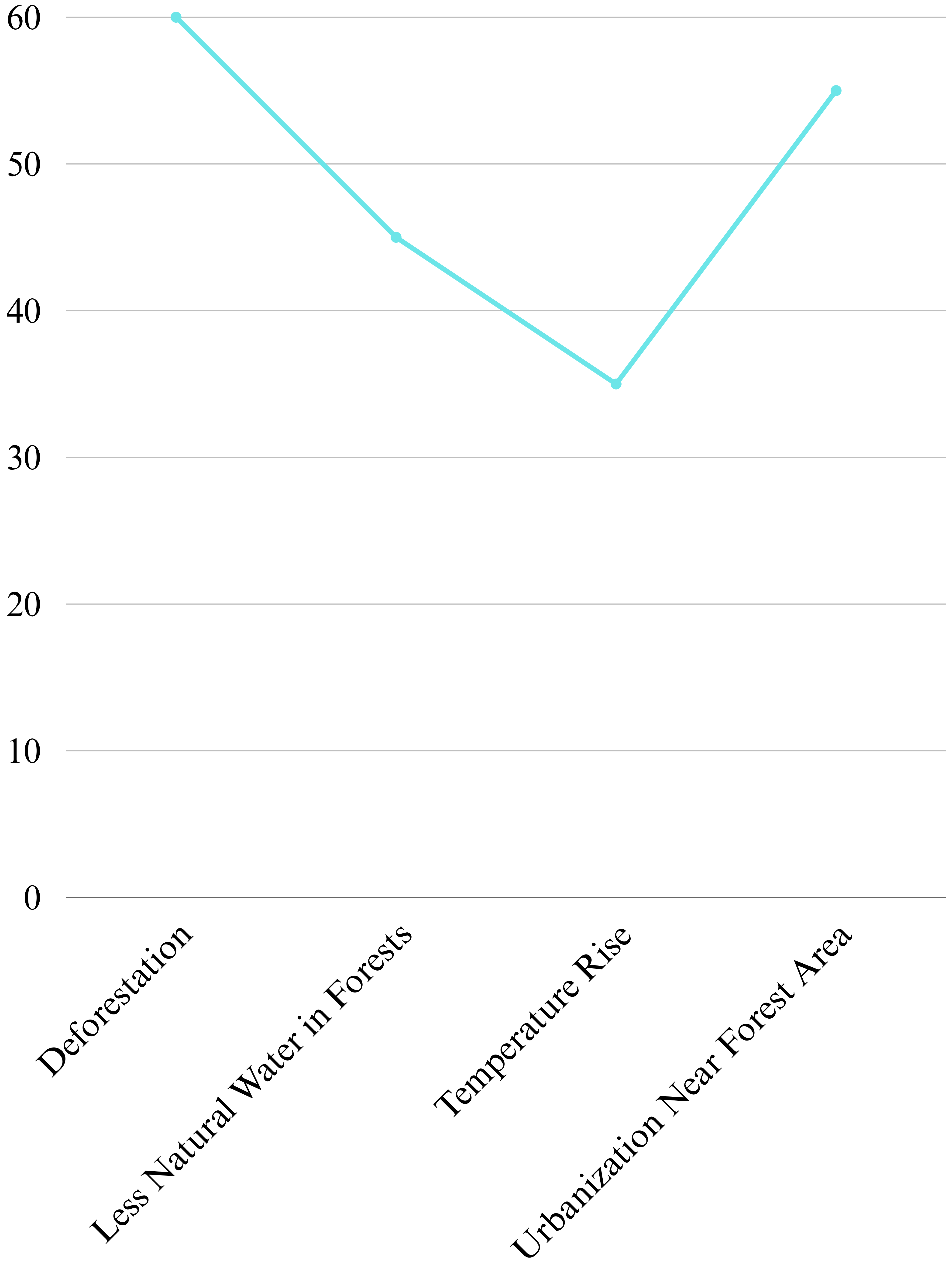
The chart shows that most wild animal intrusions happen near field boundaries and inside crop areas. Forest edges and water sources are also common entry points. Fewer incidents occur near storage and animal shelters.

Technologies Used in Farms

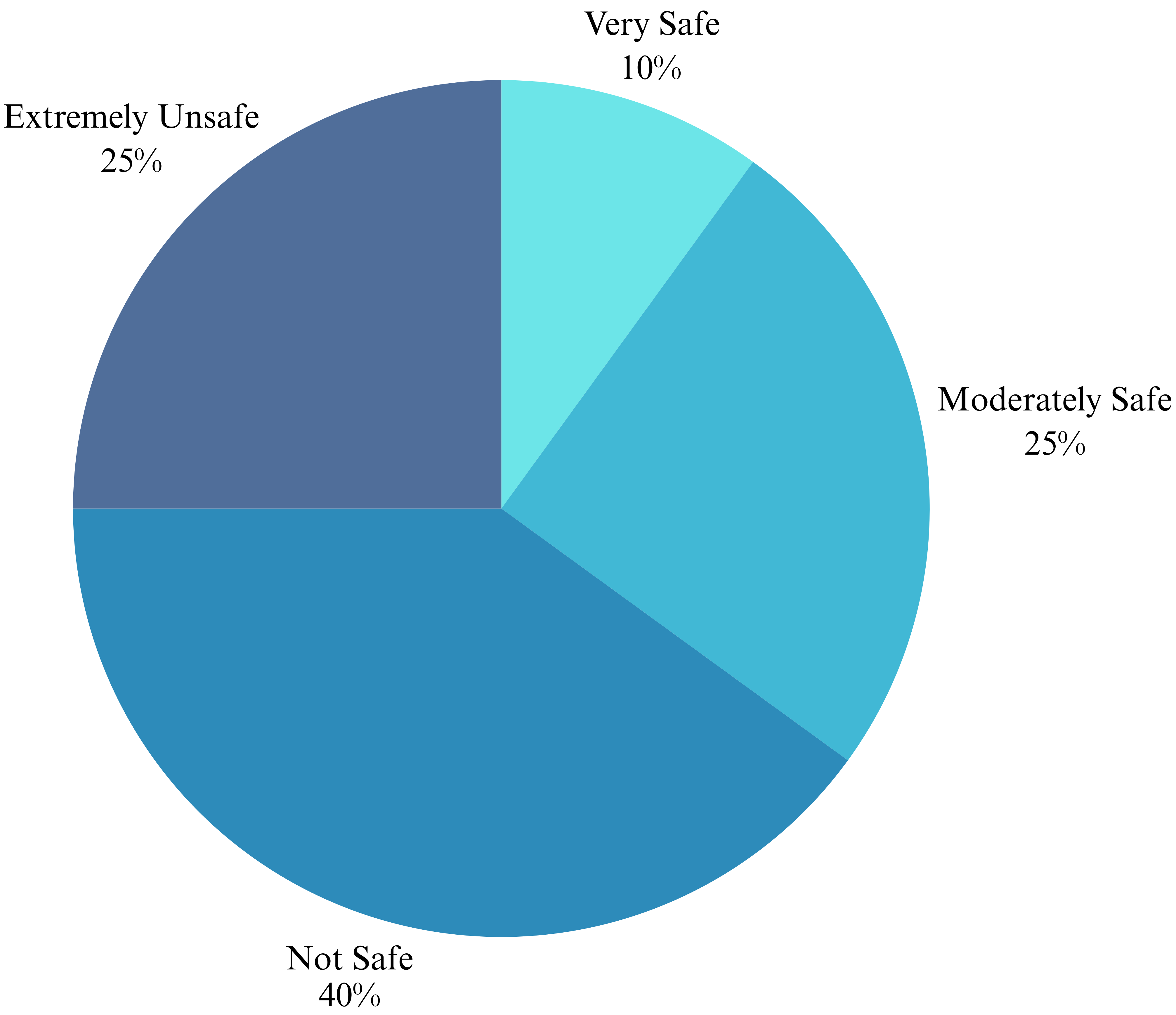


Half of the farmers still rely on manual patrolling, while modern solutions like sound-based devices are used by only 10%. This gap highlights the need for affordable and effective tech like Carronade to protect farms.

Climate Change Increases Farm Intrusion

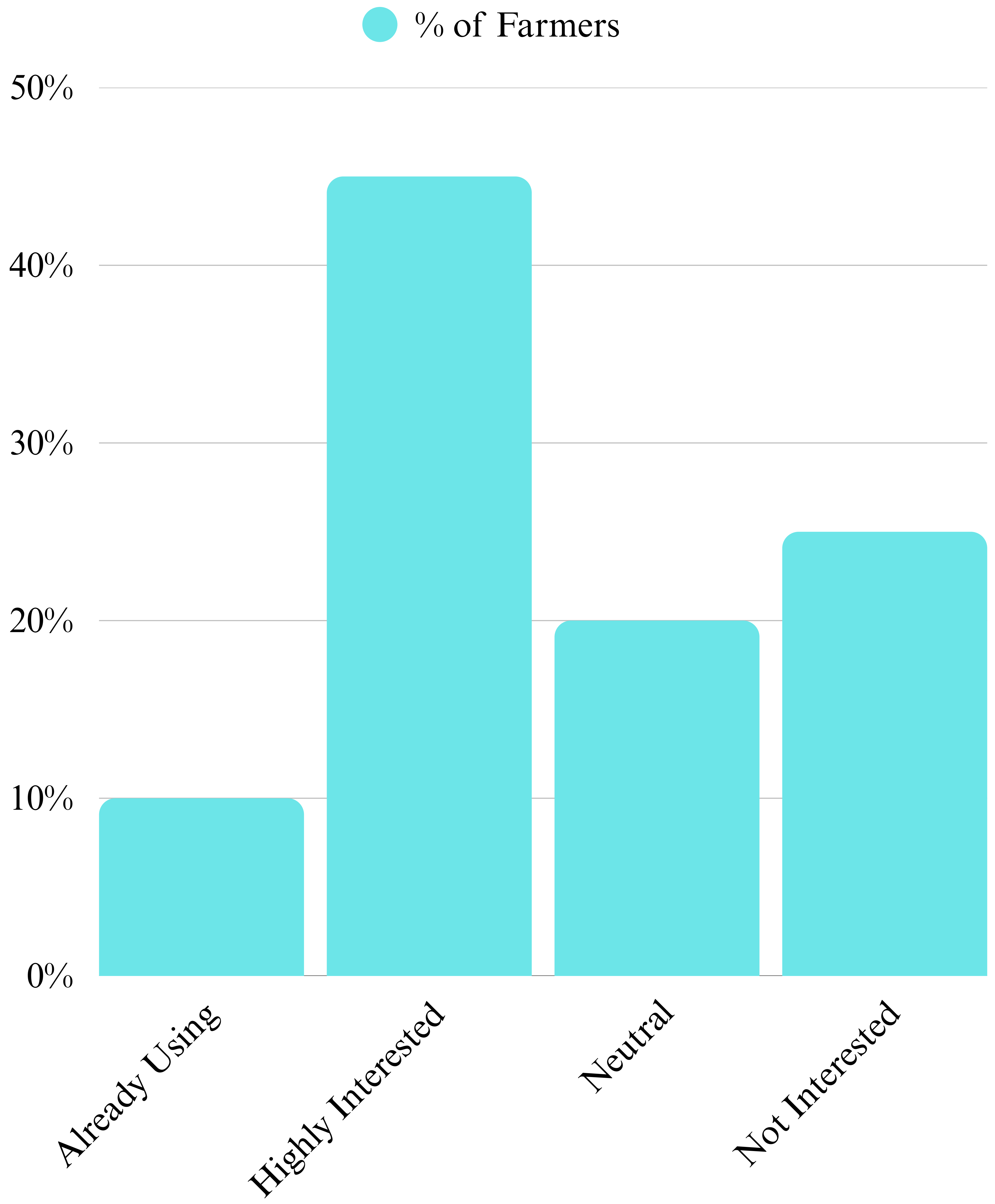


How Protected Are Our Crops?



Only 10% of farmers feel truly safe, while 65% feel moderately or extremely unsafe. This reveals a lack of reliable security measures and the constant fear farmers live with.

Sound-Based Repellents in Modern Farming



CASE STUDIES

Case Study 1: Rajendra Singh (Udaipur, Rajasthan)

Rajendra Singh is a wheat farmer based in the rural outskirts of Udaipur, Rajasthan. His 3-acre field was frequently targeted by wild boars, especially at night, causing extensive crop damage and financial losses. Rajendra tried traditional methods like fencing and scarecrows, but nothing worked effectively.

In February 2025, he installed a solar-powered Carronade device near the border of his field. The Carronade emitted high-frequency sound waves that startled and deterred wild boars without harming them. Over the next two weeks, Rajendra noticed an 80% reduction in crop damage. The Carronade's automatic activation system during night hours proved to be a reliable and low-maintenance solution, allowing Rajendra to sleep peacefully and reduce additional labor costs.

Case Study 2: Meera Devi (Haldwani, Uttarakhand)

Meera Devi owns a 2-acre fruit orchard in Haldwani where she grows guavas and lychees. Every morning, troops of monkeys raided her farm, damaging a significant portion of the produce. She faced seasonal losses and had to employ labor just to shoo the animals away.

In March 2024, Meera installed a Carronade unit near the orchard's entry point. Night Camera made it easy to monitor in night and protect the farm. Within a month, monkey raids were reduced by over 70%. The Carronade, being solar-powered and weather-resistant, required minimal upkeep. It allowed Meera to complete her harvest cycle uninterrupted, resulting in increased yield and profits.

Case Study 3: Sukhdev Patel (Rewa, Madhya Pradesh)

Sukhdev Patel, a vegetable farmer from Rewa, was experiencing consistent crop damage due to deer entering his field at night. The losses amounted to nearly ₹10,000 every month. Traditional fencing methods failed due to the deer's ability to leap over barriers.

In January 2025, Sukhdev opted for Carronade with a motion-sensor feature. Every time deer approached, the device activated an ultrasonic alarm. This startled the animals and deterred them from returning and alarming him to make a boom. Within just a few weeks, Sukhdev observed that deer no longer ventured near the field. With an effective protection range covering over 1 acre, Carronade became a cost-effective and sustainable solution for his farm.

Case Study 4: Kamla Bai (Jhalawar, Rajasthan)

Kamla Bai is an elderly farmer managing a grain crop field in Jhalawar. Her biggest challenge was birds pecking on crops during the ripening stage, especially sparrows and mynas. These birds would eat up to 20% of her total yield, impacting her yearly earnings.

In December 2024, Kamla installed Carronade configured for high-pitch bird-repelling frequencies. Positioned near the crop's center, the device worked throughout daylight hours. Within a harvesting cycle, she noticed over a 60% increase in yield due to reduced bird interference. Kamla praised the device's ease of use and found it much safer than using nets or chemical repellents.

Case Study 5: Laxman Reddy (Guntur, Andhra Pradesh)

Laxman Reddy grows tomatoes and green chilies on his 5-acre farm. Monkeys and stray dogs frequently trampled his vegetables, resulting in significant losses. Daytime guards were expensive, and fencing was not practical for such a large area.

In November 2024, Laxman installed two Carronade devices—one near the main gate and the other in the midfield area. Within one week, the number of monkey entries dropped to zero, and dog intrusions became rare. The Carronade's sound feature kept animals away from the farm. As a result, his vegetable yield improved significantly, and he could cut down labor costs.

Case Study 6: Harpreet Kaur (Ludhiana, Punjab)

Harpreet Kaur manages a family-owned vegetable farm in Ludhiana. With children often playing nearby, she was hesitant to install electric fences due to safety concerns. She needed a safer, animal-friendly option.

In July 2024, she chose Carronade for its noise-based deterrent system. The solar-powered device was installed near the field boundary and programmed to activate at sunrise and sunset. Within a month, Harpreet reported zero incidents of stray animals damaging her crops. Most importantly, the device provided a sense of security for her family while being environmentally friendly.

Case Study 7: Shamsher Khan (Sikar, Rajasthan)

Shamsher Khan’s farm is situated on hilly terrain where building physical barriers like fences is impractical and expensive. Wild boars and jackals often invaded his land, uprooting tuber crops.

In December 2024, he installed two Carronade units along the lower and upper slopes. The devices covered wide angles and made a sonic boom to scare the animals. Within a few weeks, wild animal activity dropped dramatically. Shamsher now considers the Carronade a must-have for any farmer dealing with difficult terrain.

Case Study 8: Devika Nair (Mumbai)

Devika Nair grows rice in her field, where occasional bird movement poses a serious threat. While her farm wasn’t frequently targeted, even a single flock of birds' visit could destroy months of work.

In October 2024, she installed Carronade along with alarming camera module . While birds were rare visitors, the warning system activated at a 30-meter range, preventing any close encounters. Devika also coordinated with nearby farms to install these systems. Since then, no birds have entered the area, ensuring the safety of both crops and workers.

Case Study 9: Nikhil Joshi (Nashik, Maharashtra)

Nikhil Joshi used traditional scarecrows to protect his grape vineyard from monkeys. Unfortunately, the monkeys became accustomed to them, and damage continued. Frustrated, he looked for an alternative.

In March 2024, he installed a Carronade with a motion detection module. This ensured that he gets notification of intrusions. After three weeks, he noticed a 90% drop in monkey visits. His vineyard showed improved growth and quality, and he recommended Carronade to other local farmers.

Case Study 10: Savita Sharma (Bilaspur, Chhattisgarh)

Savita Sharma managed a medium-sized farm and was dependent on night-time laborers to keep watch over the crops, costing her ₹2,000–₹3,000 per week. She needed an affordable and automated deterrent system.

In November 2024, she opted for Carronade with motion detection and solar charging. It activated instantly upon animal movement and covered a wide area. Within the first month, she completely stopped hiring guards, saving over ₹15,000 annually. The device also gave her peace of mind during nighttime and became an integral part of her farm’s defense system.

PROPOSED SOLUTION

Our solution, Carronade, is an automated, eco-friendly crop protection system that uses a smart combination of chemical reaction and motion sensing to deter pests. It operates by reacting calcium carbide with water to produce acetylene gas, which is then ignited to create a loud, non-lethal sound that scares away birds, animals, and locusts.

Carronade is a smart, automated crop protection system designed to safeguard agricultural fields from birds, locusts, and stray animals through a non-lethal and environmentally friendly approach. Traditional methods of pest control—such as scarecrows, chemical sprays, and manual guarding—often fall short in effectiveness, especially at night or over long periods. Carronade addresses these challenges by combining **motion-sensing technology, a controlled chemical reaction, and intelligent automation** in one integrated system.

The core working principle of Carronade is based on the **chemical reaction between calcium carbide and water**, which produces acetylene gas. This gas is then ignited using an electric spark to generate a **loud explosive sound that repels intruders without causing them harm.** What sets Carronade apart is its use of a **motion-detecting night-vision camera** to activate the system only when intruders are detected nearby. This smart activation conserves energy and reduces unnecessary noise pollution.

Carronade is constructed with **durable metallic materials**, making it suitable for continuous outdoor use under various weather conditions. Its components are **affordable and designed for easy maintenance**, making it an ideal solution for small to medium-scale farmers, particularly in rural areas.

FEATURES

- **Motion Detection with Night Vision:** The Carronade is equipped with a sophisticated camera system that can **detect movement** in both bright daylight and low-light conditions. This advanced feature ensures that the system is **activated only when pests & intruders** are present, allowing for selective and efficient operation.
- **Eco-Friendly Acoustic Deterrent:** Unlike traditional pest control methods that rely on harmful chemicals or traps, the **Carronade uses sound to scare away pests & intruders**. This approach is completely non-lethal, making it an excellent choice for farmers as it doesn’t harms the animals also.
- **Calcium Carbide–Based Sound Mechanism:** Carronade has a servo motor that dispenses calcium carbide into a **specially designed reaction chamber**. A timed water pump then adds water to this chamber, triggering a chemical reaction that produces acetylene gas. This gas is subsequently ignited by an electric spark, **resulting in a loud bang that effectively repels pests.**
- **Dual-Stage Ignition System:** To ensure both safety and effectiveness, the Carronade features a **dual-stage ignition system**. The **first spark is used to clear any residual gas** from the chamber, while the s**econd spark ignites the freshly formed acetylene**, ensuring consistent and reliable sound generation.

FEATURES

- **Melem MOF Sheets for Gas Absorption:** To **prevent environmental contamination and enhance safety**, the Carronade incorporates Melem MOF sheets at the gas outlet. These sheets **absorb any excess acetylene gas**, ensuring that **no harmful substances are released** into the environment.
- **Automated Control and Logic Programming:** The Carronade is controlled by a microcontroller which manages each step in the cycle with **precision and reliability**. This automated control ensures that the system operates **smoothly and efficiently.**
- **Energy Efficient and Low Power Consumption:** Carronade is highly energy efficient. It is suitable for solar or battery power, making it ideal for remote rural applications where access to electricity may be limited.
- **Cost-Effective and Scalable:** Built using **low-cost materials and widely available components**, the Carronade is an **affordable solution** for pest control. It is **easily scalable**, making it suitable for both small and large farms, allowing farmers to adapt the system to their specific needs.

METHODOLOGY

The development and functioning of Carronade involve a multidisciplinary approach that integrates principles of chemical engineering, mechanical actuation, electronic automation, environmental science, and agricultural technology. The methodology outlines how various components are designed, assembled, and function in a cohesive automated cycle to deter birds, locusts, and stray animals from damaging crops.

The process includes:

- Material selection
- Hardware integration
- Sensor calibration
- Software development
- System synchronization
- Field deployment
- Testing and validation

MATERIALS AND COMPONENT SPECIFICATION

Structural Materials

- Main Frame: Galvanized iron (GI) or stainless steel (SS304) for corrosion resistance and durability in harsh weather.
- Combustion Chamber: Heat-resistant stainless steel (SS316).
- Protective Casings: IP65-rated metal enclosures for electronics and sensors.

METHODOLOGY

Chemical Components

- Calcium Carbide (CaC_2):
  - Industrial-grade granules.
  - Stored in a sealed metal container with moisture protection.
- Water Source:
  - Stored in a corrosion-resistant tank.
  - Delivered through PVC tubing.
- Melem MOF Powder:
  - Synthesized Melem-based Metal–Organic Framework.
  - Compressed into sheets and placed at the gas outlet to adsorb unreacted acetylene gas (C_2H_2), reducing environmental impact.

SYSTEM DESIGNING AND INTEGRATION

Motion Detection Camera

- Camera Type: Low-light night-vision camera with PIR (Passive Infrared) motion detection capability.
- Mounting: Vertically mounted for wide field-of-view
- Sensor Parameters:
  - Motion threshold: Configurable via firmware (e.g., sensitivity level 1–10).
- Activation Logic:
  - Image processing + infrared trigger.
  - Filters false positives from wind or moving vegetation.

METHODOLOGY

Dispensing Subsystem

- Calcium Carbide Container:
  - Metal container with servo-controlled gate.
  - Dispenses ~5–10 grams per cycle depending on field size.
  - Positioned to release directly into combustion chamber.
- Water Storage Tank:
  - Low-voltage submersible pump (12V DC).
  - Ensures controlled reaction intensity.

Combustion and Ignition System

- Dual Spark System:
  - Spark 1 (Pre-ignition): Clears residual gas from previous cycle.
  - Spark 2 (Main ignition): Triggers the controlled explosion.
- Combustion Chamber:
  - Volume: ~850-1250 mL.
  - Equipped with MOF Sheets.

APPLICATIONS

1. Agricultural Crop Protection

Carronade is primarily used to deter wild animals such as monkeys, wild boars, deer, and nilgai from destroying crops. Its sonic boom acts as a non-lethal, eco-friendly solution that safeguards produce without harming animals or the environment.

2. Forest Fringe Villages

In villages located near forests and wildlife reserves, Carronade can prevent wild animal intrusions into homes and farmlands. It minimizes human-wildlife conflict while ensuring both human safety and animal conservation.

installed around cattle shelters. It ensures the safety of animals during nighttime without the need for dangerous fencing.

3. Eco-Tourism and Campsites

Carronade helps maintain a safe perimeter in jungle resorts and forest lodges by discouraging wildlife from entering tourist zones. This allows visitors to enjoy nature without the risk of surprise animal encounters.

4. Cattle Sheds and Gaushalas

To protect livestock from predators such as leopards, wolves, or wild dogs, Carronade can be installed around cattle shelters. It ensures the safety of animals during nighttime without the need for dangerous fencing.

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APPLICATIONS

5. Schools and Ashrams in Remote Areas

Remote schools and spiritual institutions often face nuisance from monkeys and stray animals. Carronade provides a safe and quiet solution to keep such premises secure and hygienic, especially for children and residents.

6. Railway Tracks and Road Crossings

Carronade can be deployed at sensitive points along railway lines or forest roads to prevent animals from crossing and getting hit. This improves safety for both wildlife and commuters, especially in regions with frequent animal movement.

system

7. Home Gardens and Rooftop Farms

Urban gardeners can use smaller Carronade units to protect rooftop farms or kitchen gardens from birds, squirrels, and stray cats. It supports urban farming efforts by preserving produce in a non-harmful way.

8. Disaster Response and Emergency Zones

In emergency situations like floods or forest fires, Carronade can be used to redirect or guide animals away from human settlements or relief camps. It aids in both human and animal protection during crises.

irrigation

MARKET POTENTIAL

The Carronade is well-positioned to tap into a rapidly growing demand for eco-friendly, automated, and cost-effective solutions in the global agriculture sector. It addresses key challenges of crop protection against pests—particularly birds, locusts, and stray animals—by offering a non-lethal alternative to traditional methods like pesticides, scarecrows, and manual guarding. With growing concerns about environmental sustainability and food security, the Carronade's eco-friendly and automated design provides a scalable solution for various agricultural settings.

- Global Pest Control Market: The pest control industry is worth over $80 billion globally and is expected to grow at a 5-7% CAGR driven by an increasing focus on sustainable and organic farming methods.
- Smart Agriculture and Automation: The demand for smart farming solutions is expanding, with the agriculture automation market projected to reach $16.2 billion by 2025. Carronade fits well into this trend, providing an automated, energy-efficient solution.
- Environmental Concerns: With increasing regulations on harmful chemical usage, there is an escalating demand for safe and non-toxic pest deterrents. Carronade, using a sound-based mechanism and chemical reaction, addresses this need perfectly.

MARKET POTENTIAL

Feasibility

- Low-cost, scalable design suitable for all farm sizes
- Automated operation reduces labor and manual guarding
- Solar/battery compatible, ideal for remote areas
- Eco-friendly and safe, using Melem MOF sheets to neutralize gas

Target Market

- Small & medium farmers in rural and semi-urban areas
- Organic farms needing chemical-free pest control
- Large farms seeking automation
- Emerging markets (Asia, Africa, Latin America)
- Government/NGO agricultural programs